



LEAF SPRING ANALYSIS

Mihriban Orman 201615041

Mechanical Engineering Department, Cankaya University, Ankara, Turkey

ABSTRACT

In this report, I modeled the leaf spring length parametrically. With the ANSYS program, I analyzed the total deformation and stress change by changing the shape of the leaf spring.

Keywords: ANSYS, total deformation, stress, leaf spring

NOMENCLATURE

σ	stress
F	Force
ε	strain
L	length
A	Cross sectional area
E	Elastic modulus of material

1. INTRODUCTION

In this report, I modeled the leaf spring length parametrically. With the ANSYS program, I analyzed the total deformation and stress change by changing the shape of the leaf spring. In the first part, I drew the shape on the ANSYS and selected some lengths as parameters. Then I chose the mesh size for this drawn geometry and the maximum number of meshes allowed by my student license. After finishing knitting, I went to set-up. In the set-up section, I fixed the motion in other directions, allowing the spring to rotate only in the x direction and move in the z direction. I applied the force in the middle of the spring and analyzed the results. In the entire analysis units are written in millimeters and newtons.

2. MATERIALS AND METHODS

Starting from the geometry, all the parties of the analysis are as follows;

2.1.Geometry:

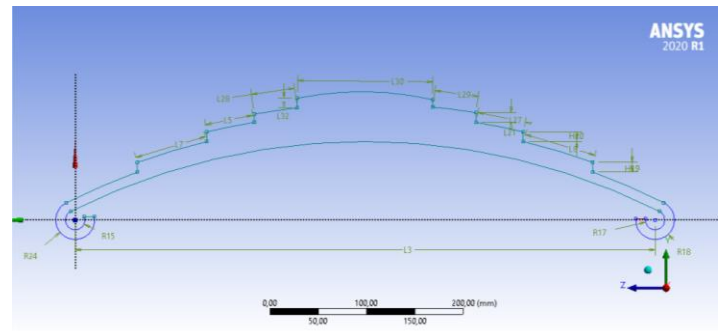


Figure: Drawn leaf spring and length parameters

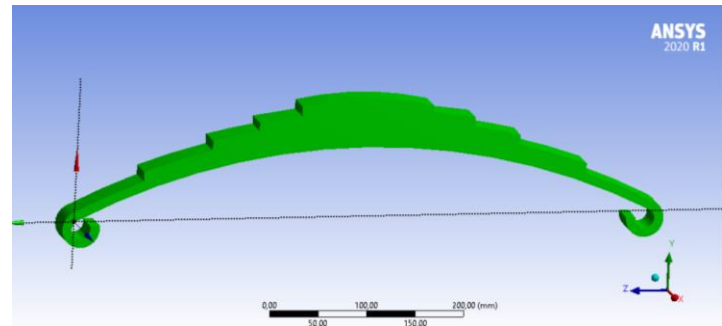


Figure: Shape of the extruded leaf spring

Details View	
Details of Sketch3	
Sketch	Sketch3
Sketch Visibility	Hide Sketch
Show Constraints?	No
Dimensions: 16	
P H19	10 mm
P H20	10 mm
P L21	10 mm
L27	50 mm
L28	45 mm
L29	45 mm
P L3	600 mm
L30	140,17 mm
P L32	10 mm
L5	50 mm
L7	75 mm
L8	75 mm
P R15	10 mm
P R17	10 mm
P R18	20 mm
P R24	20 mm
Edges: 24	
Circular Arc	Cr7
Circular Arc	Cr8
Circular Arc	Cr9

Tree Outline	
[-] A: parameter shape	
XYPlane	
ZXPlane	
YZPlane	
Sketch3	
Extrude1	
Sketch3	
1 Part, 1 Body	
Solid	

Details View	
Details of Body	
Body	Solid
Volume	...
Surface Area	...
Faces	26
Edges	72
Vertices	48
Fluid/Solid	Solid
Shared Topology Method	Automatic
Geometry Type	DesignModeler

Figure: Details of the solid model formed

Figure: Length of leaf spring and selected parameters

Tree Outline	
[-] A: parameter shape	
XYPlane	
ZXPlane	
YZPlane	
Sketch3	
Extrude1	
Sketch3	
1 Part, 1 Body	
Solid	

Sketching	
Details View	
Details of Extrude1	
Extrude	Extrude1
Geometry	Sketch3
Operation	Add Material
Direction Vector	None (Normal)
Direction	Normal
Extent Type	Fixed
P FD1, Depth (>0)	30 mm
As Thin/Surface?	No
Merge Topology?	Yes
Geometry Selection: 1	
Sketch	Sketch3

Figure: Selecting the shape as the extured length and parameter

Outline	
Name	Search Outline
Project	
Model (A4)	
Geometry	
Materials	
Structural Steel	
Structural Steel 2	
Structural Steel 5	
Structural Steel 6	
Structural Steel 7	
Structural Steel 8	
Structural Steel 4	
Coordinate Systems	
Mesh	
Face Sizing	
Static Structural (AS)	
Analysis Settings	
Force	
Remote Displacement	
Remote Displacement 2	

Details of "Materials"	
Statistics	
Materials	8
Material Assignments	0

Figure : Shows material of leaf spring

2.2. Meshing:

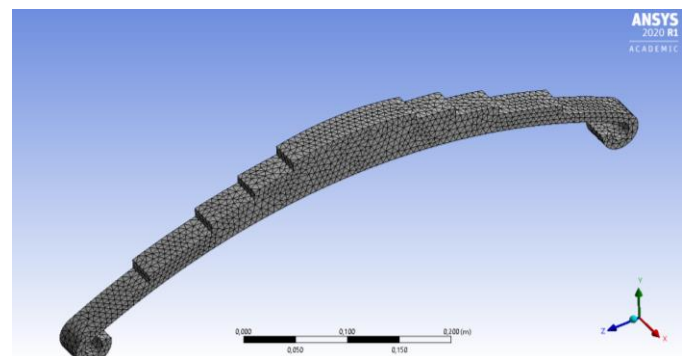


Figure: Mesh applied to the created model

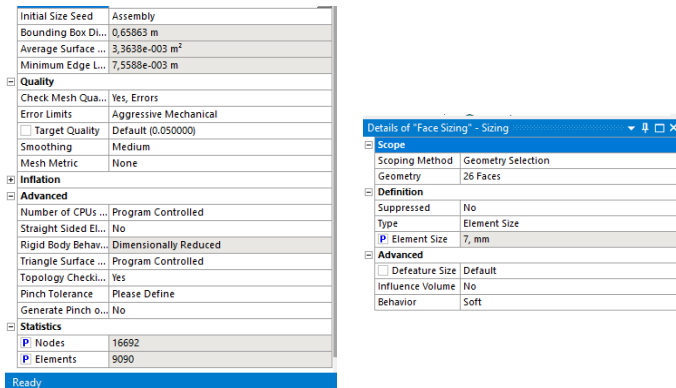


Figure: Details of mesh and mesh sizing which is selected as parameter

Comment: I chose mesh as paramter. By changing the mesh size, I field the mesh sensitivity. Mesh type is element size.

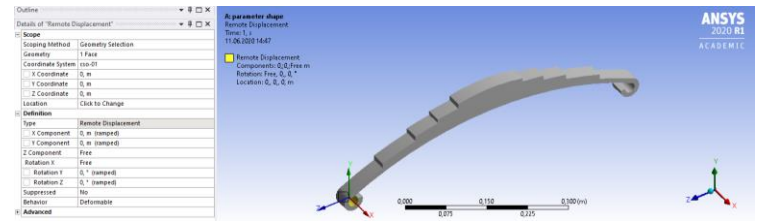


Figure : Details and location of the left support fixed

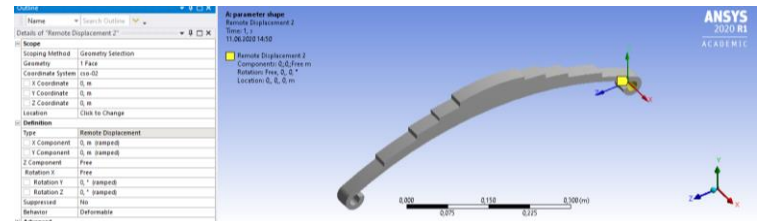


Figure : Details and location of the right support fixed

2.3. Solution:

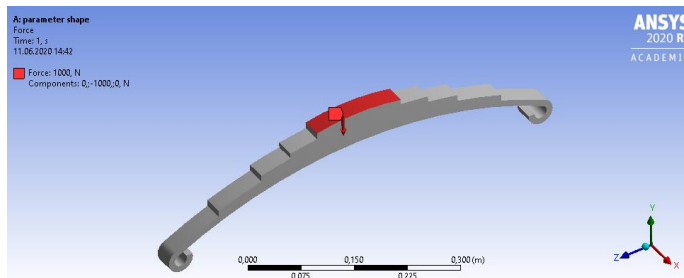


Figure: Position and direction of force relative to leaf spring

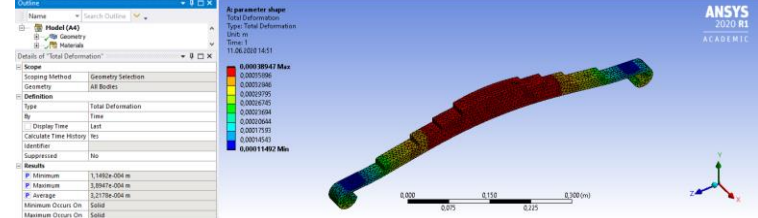


Figure: Total deformation image and selection of values as parameters for main parameter

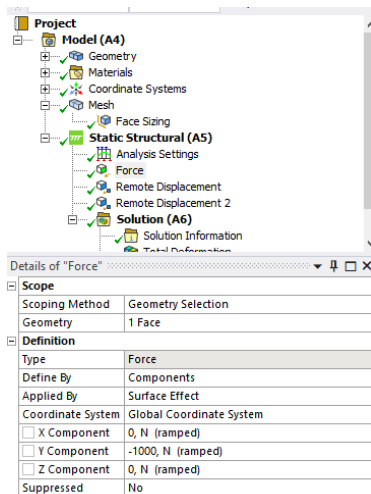


Figure: Details and magnitude of force relative to leaf spring

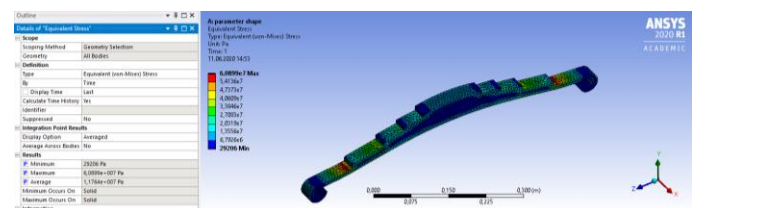


Figure: Image of stress and selection of values as parameters for main parameter

2.3. Formulas:

2.3.1. Stress:

$$\sigma = \frac{F}{A}$$

F= force applied

A=cross sectional area where force is applied

σ = stress

2.3.2. Deformation

$$\delta = \frac{FL}{EA}$$

E= elastic modulus of material (structural steel)

F= force applied

A= cross sectional area where force is applied

L= the original length of the fiber the original length of the spring

Hand calculation for main shape:

F= 1000 N

L= 600 mm

A = (140 mm) x (30mm) = 4200 mm²

E= 210000 MPa

$$\sigma = \frac{1000 \text{ N}}{4200 \text{ mm}^2} = 0.2381 \text{ MPa}$$

$$\delta = \frac{1000 \text{ N} \times 600 \text{ mm}}{21 \times 10^4 \text{ MPa} \times 4200 \text{ mm}^2} = 0,00068027 \text{ mm}$$

3. RESULTS AND DISCUSSION

Analysis of parameters:

Table of Design Points																			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P			
1	Name	Units	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14			
2	Units																		
3	SP 0 Elements	1	600	20	30	10	10	7	0.28947	0.11402	0.10270	40.889	0.07009	11.764	0.0002	0.000			
4	SP 1	2	600	20	30	10	10	7	0.23346	0.049011	0.10607	36.500	0.117941	7.2463	0.0004	0.0004			
5	SP 4	3	600	20	30	10	10	7	0.7946	0.049012	2.271	40.176	0.11124	26.498	0.0173	0.0000			
6	SP 5	4	600	20	30	10	10	7	9.9	0.1033	260.86	0.03027	46.262	0.0076	0.0000				
7	SP 6	5	1000	20	30	10	10	7	41.867	2.873	360.17	0.03044	87.603	0.0003	0.0000				
8	SP 10	6	600	20	30	11	11	7	0.00076	0.00008	9.4402	76.433	0.03043	11.131	0.0011	0.0004			
9	SP 11	12	600	20	30	12	12	7	0.62703	0.000714	0.49201	96.712	0.04010	12.875	0.0020	0.000			
10	SP 12	13	600	20	30	13	13	7	0.00076	0.00008	0.00000	236.2	0.01117	12.827	0.0011	0.000			
11	SP 13	14	600	20	30	15	15	7	1.4912	0.17919	1.0982	242.26	0.03076	26.89	0.013	0.007			
12	SP 14	15	600	20	30	16	16	7	0.00076	0.00008	0.00000	26.867	0.010667	6.046	0.001	0.000			
13	SP 15	16	600	20	30	18	18	7	0.00076	0.00008	0.00000	40.889	0.02006	11.764	0.0002	0.000			
14	SP 16	17	600	20	30	18	18	8	0.4171	0.000704	0.00000	61.676	0.02061	11.646	0.0003	0.000			
15	SP 17	18	600	20	30	18	18	8	0.00076	0.00008	0.00000	59.439	0.01004	11.5	0.001	0.000			
16	SP 18	19	600	20	30	18	18	10	0.00076	0.00008	0.00000	59.57	0.00000	11.003	0.000	0.000			
17	SP 19	20	600	20	30	18	18	11	0.00076	0.00008	0.00000	60.127	0.00001	11.508	0.000	0.000			
18	SP 20	21	600	20	30	18	18	12	0.00076	0.00008	0.00000	59.091	0.00072	11.508	0.007	0.000			
19	SP 21	22	600	20	30	18	18	13	0.00076	0.00008	0.00000	57.14	0.00000	11.521	0.000	0.000			
20	SP 22	23	600	20	30	18	18	14	0.00076	0.00008	0.00000	57.462	0.00033	11.404	0.004	0.000			
21	SP 23	24	600	20	30	18	18	15	0.00076	0.00008	0.00000	56.615	0.00038	11.322	0.005	0.000			
22	SP 24	25	600	20	30	18	18	16	0.00076	0.00008	0.00000	55.889	0.00038	11.764	0.000	0.000			
23	SP 25	26	600	20	30	18	18	17	0.00076	0.00008	0.00000	49.324	0.01440	6.6761	0.0013	0.000			
24	SP 26	27	600	20	30	18	18	18	0.00076	0.00008	0.00000	13.071	0.00000	6.176	0.000	0.000			
25	SP 30	31	600	20	30	18	18	7	0.13111	0.001196	0.12007	26.890	0.00100	6.000	0.000	0.000			
26	SP 31	32	600	20	30	18	18	7	0.14643	0.001007	0.00000	26.206	0.00000	6.1317	0.0017	0.000			
27	SP 32	33	600	20	30	18	18	7	0.00076	0.00008	0.00000	15.761	0.00000	6.1124	0.000	0.000			
28	SP 33	34	600	20	30	18	18	7	0.00076	0.00008	0.00000	11.768	0.00000	5.889	0.000	0.000			

Figure: Changing selected parameters separately while keeping other values constant and solutions

Comment: The important point in parametric changes is that we

If we want to investigate the effect of one parameter on the analysis, we must leave the other parameters constant while changing the parameter we select.

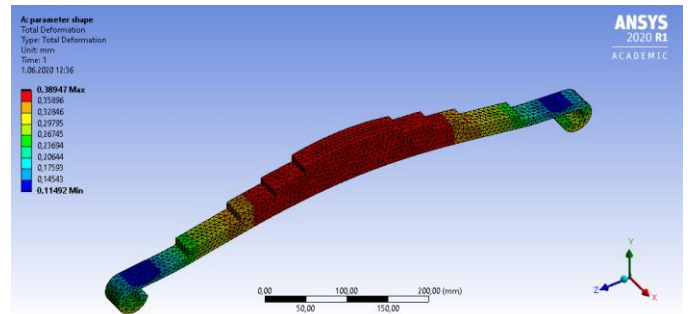


Figure: the shape of the total deformation result of the main shape

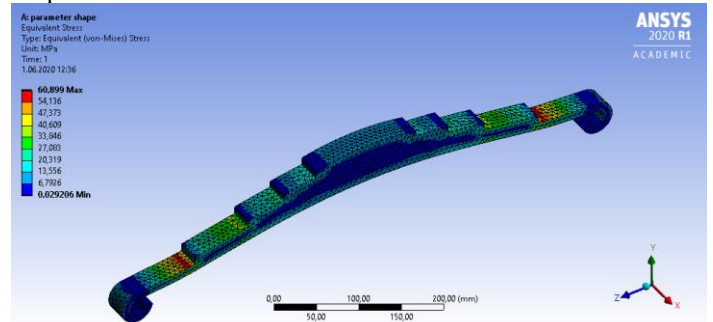


Figure: the shape of the equivalent stress result of the main shape

Mesh analysis:

P25- Face Sizing Element Size mm	mesh nodes	mesh element s	P16- Total Deforma tion Maximu m mm	P17- Total Deforma tion Minimu m mm	P18- Total Deforma tion Averagem mm	P19- Equivale nt Stress Maximu m MPa	P20- Equivale nt Stress Minimu m MPa	P21 - Equivale nt Stress Average MPa
7	16692	9090	0,39	0,11	0,32	60,90	0,03	11,76
7	16692	9090	0,39	0,11	0,32	60,90	0,03	11,76
8	11328	6078	0,48	0,00	0,36	61,68	0,03	11,65
9	9177	4832	0,39	0,12	0,32	59,42	0,06	11,50
10	6699	3420	0,39	0,11	0,33	59,57	0,06	11,10
11	5630	2858	0,52	0,03	0,38	60,18	0,09	11,37
12	4867	2441	0,40	0,04	0,33	59,09	0,06	11,40
13	4388	2189	0,39	0,05	0,33	57,54	0,07	11,52
14	4114	2040	0,39	0,07	0,33	57,40	0,10	11,40
15	3275	1581	0,38	0,08	0,32	59,62	0,04	11,82

Figure: Table of mesh sensitivity

Comment: The important point in network analysis is to make the size of the network as small as possible. The smaller network size gives us more accurate analysis.

Parameters of different support thickness:

P11 - radius inner radii mm	P14 - mm	P22 - Mesh Nodes	P23 - Mesh Elements	P16 - Total Deformation Maximum mm	P17 - Total Deformation Minimum mm	P18 - Total Deformation Average mm	P19 - Equivalent Stress Maximum MPa	P20 - Equivalent Stress Minimum MPa	P21 - Equivalent Stress Average MPa
20	10	16692	9090	0,389	0,115	0,322	60,899	0,029	11,764
22	10	16411	8898	0,296	0,025	0,243	44,324	0,015	9,679
24	10	16960	9228	0,228	0,023	0,185	32,071	0,007	8,178
28	10	19643	10759	0,158	0,001	0,120	20,958	0,002	6,101
30	10	20137	11080	0,125	0,022	0,095	18,236	0,005	5,318
35	10	22081	12214	0,097	0,000	0,069	15,701	0,004	4,013
40	10	24407	13520	0,068	0,000	0,047	11,758	0,004	3,086

Figure: table of parameter of different support thickness

Comment: Total deformation and equivalent stress decreased as the support thickness increased.

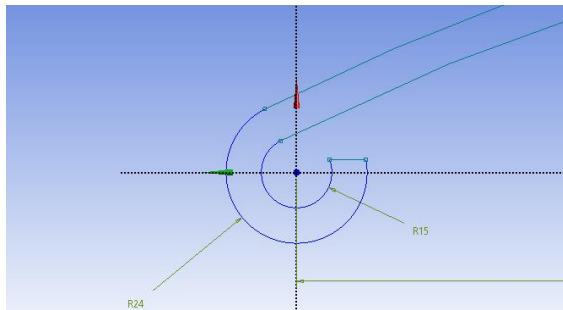


Figure: support thickness parameters

Parameters of different length shapes:

P10 - length mm	P22 - Mesh Nodes	P23 - Mesh Elements	P16 - Total Deformation Maximum mm	P17 - Total Deformation Minimum mm	P18 - Total Deformation Average mm	P19 - Equivalent Stress Maximum MPa	P20 - Equivalent Stress Minimum MPa	P21 - Equivalent Stress Average MPa
600	16692	9090	0,389	0,115	0,322	60,899	0,029	11,764
800	20171	10955	2,795	0,499	2,275	163,785	0,021	26,658
1000	23878	12888	9,900	3,272	7,815	260,959	0,030	46,252
1200	29083	15626	41,997	2,873	23,970	360,372	0,033	67,603

Figure: Table of parameter of different length

Comment: Total deformation and equivalent stress increased as the length of leaf spring increased.

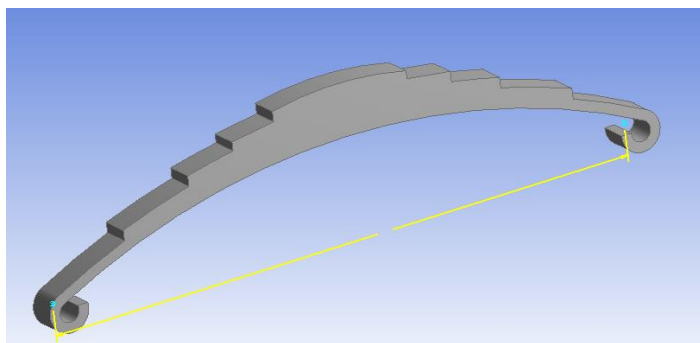


Figure: length parameter of leaf spring

Parameters of different width shapes:

P12 - width mm	P22 - Mesh Nodes	P23 - Mesh Elements	P16 - Total Deformation Maximum mm	P17 - Total Deformation Minimum mm	P18 - Total Deformation Average mm	P19 - Equivalent Stress Maximum MPa	P20 - Equivalent Stress Minimum MPa	P21 - Equivalent Stress Average MPa
30	16692	9090	0,39	0,11	0,32	60,9	0,03	11,76
50	22806	12594	0,23	0,05	0,19	36,5	0,02	7,25
60	26476	14731	0,21	0,01	0,16	29,8	0,02	6,15

Figure: Table of parameter of different width

Comment: As the width of the leaf spring increases, total deformation and equivalent stress decrease. The parameter that shows the width of the leaf spring is the extrude parameter.

4. CONCLUSION

In my analysis, when the parameters changed, total deformation and equivalent stress increased or decreased proportionally. The reason for this is that deformation and stress are directly proportional, as understood in the formula. In an ideal spring, total deformation and equivalent stress should be low. That's why:

- As the leaf spring length increased, deformation and equivalent stress increased. Therefore, leaf spring length should be short.
- As leaf spring width increased, deformation and equivalent stress decreased. Therefore, leaf spring should be wide.
- Deformation and equivalent stress decreased as the leaf spring support thickness increased. Therefore, leaf spring supports should be thick.

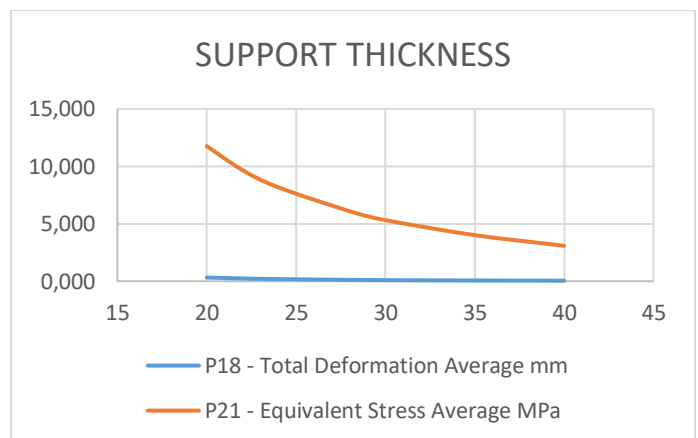


Figure: support thickness change chart

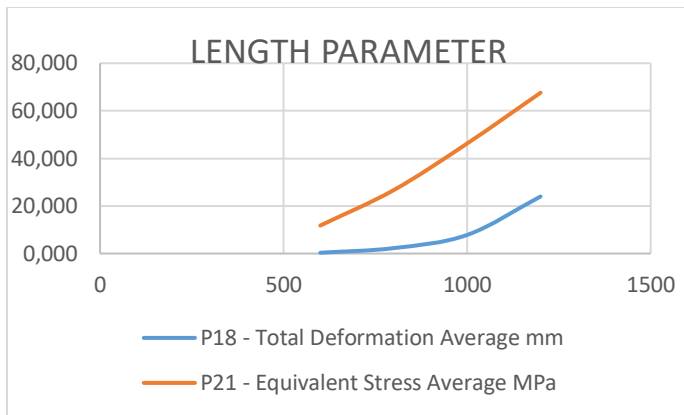


Figure: Different length change chart

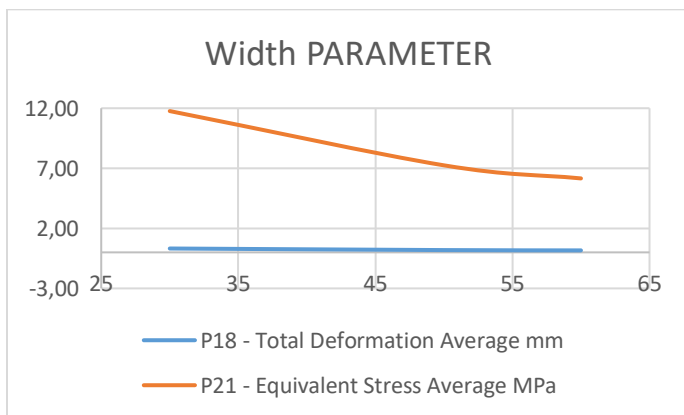


Figure: Different width change chart

REFERENCES

All information from class notes of me 402